

Lower Limb:

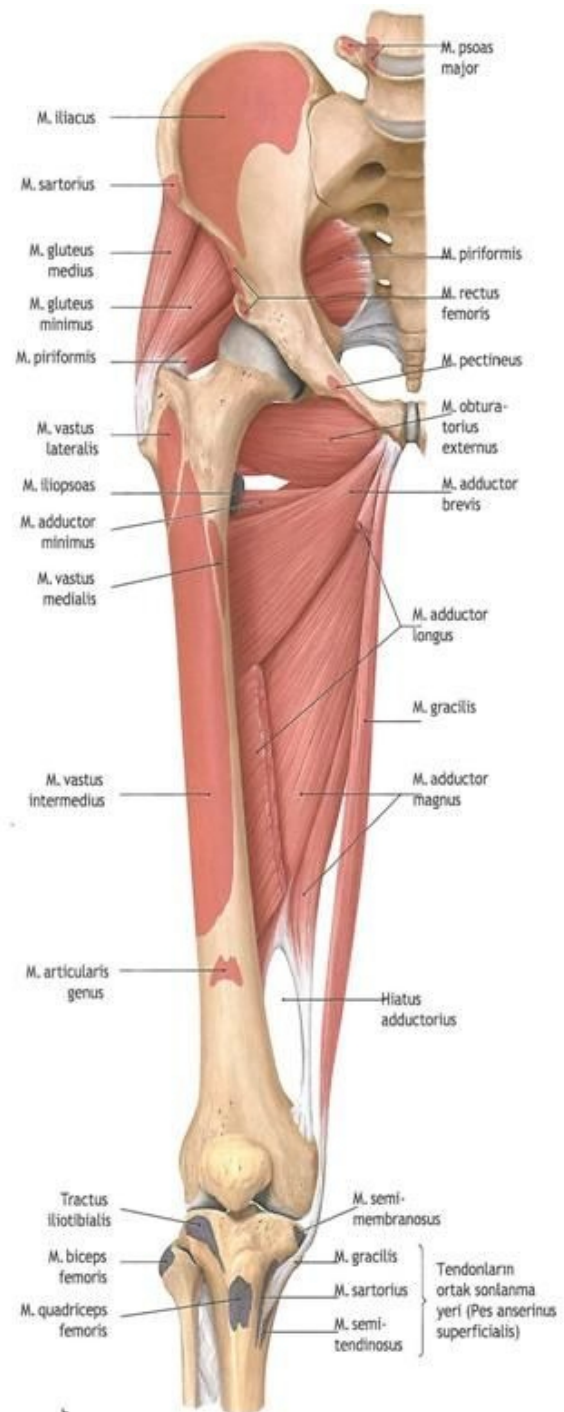
Medial Compartment of the Thigh

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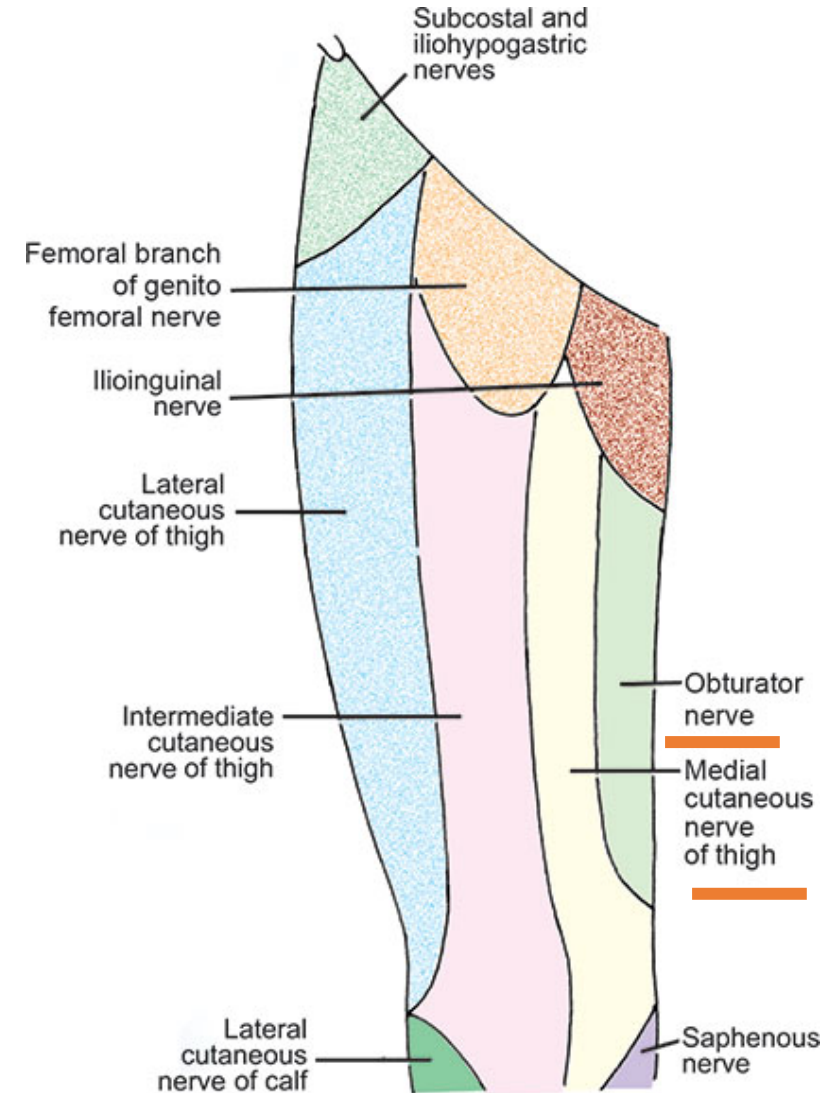
2025





Cutaneous Nerves of Medial aspect of the thigh

Cutaneous nerve	Origin	Area of skin supply in thigh
1. Medial cutaneous nerve of the thigh.	Branch of femoral nerve	To skin of Medial aspect of the thigh. - Medial part of back of thigh to the upper part of back of the leg.
2. Cutaneous branch of obturator nerve.	obturator nerve (L2,3,4)	To skin of the middle 1/3 of Medial aspect of the thigh. (region overlying adductor canal).



Content

❖ Muscles (5):

1-Gracilis

2-Adductor longus

3-Adductor brevis

4-Adductor magnus

Adductors

5-Obturator externus

❖ Blood supply: Profunda femoris artery & obturator artery.

❖ Nerve supply: Obturator nerve.

1-Gracilis

- long, strap like, lies on medial thigh, most superficial & most medial & weakest.
 - Lies on the medial side of the thigh & knee.
- O : from the outer surface of inferior pubic ramus & ramus of ischium .
- I : into upper medial surface of shaft of tibia. **(SGS)?**.
- N : Obturator nerve
- A : **hip joint**: Flexion(weak) & **Adduction**.
Knee joint: Flexion & medial rotation.

(flexor of both hip & Knee joint)



SGS on
upper
medial
aspect of
tibia

- 3 Semitendinosus
- 2 Gracilis
- 1 Sartorius

Medial head of
gastrocnemius

Soleus

Rectus femoris

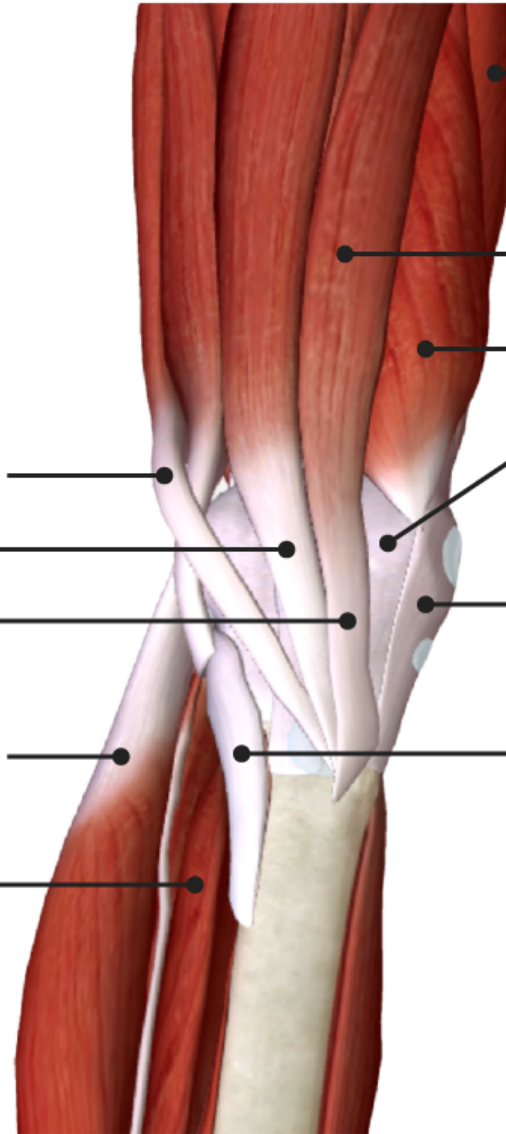
Sartorius

Vastus medialis

Fibrous joint
capsule of knee

Patellar ligament

Oblique popliteal
ligament





Upper
medial
surface of
tibia



SGS: add stability to the medial aspect of the extended knee.

- All cause flexion & medial rotation of the knee.
- Sartorius, gracilis, semitendinosus. (SGS)



3 muscles acting on both hip and knee joint;

1)Sartorius.

Flexion

Flexion

2)Rectus femoris

Flexion

Extension knee

3)Gracilis.

Flexion(weak)

Flexion

■ 2) Adductor longus

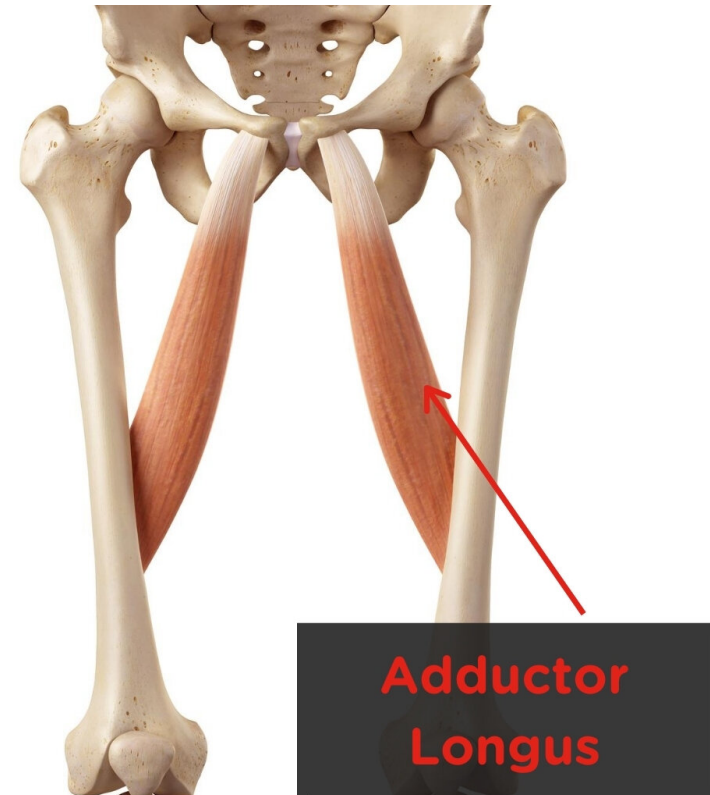
Triangular, and the most anteriorly placed of the three adductor muscles.

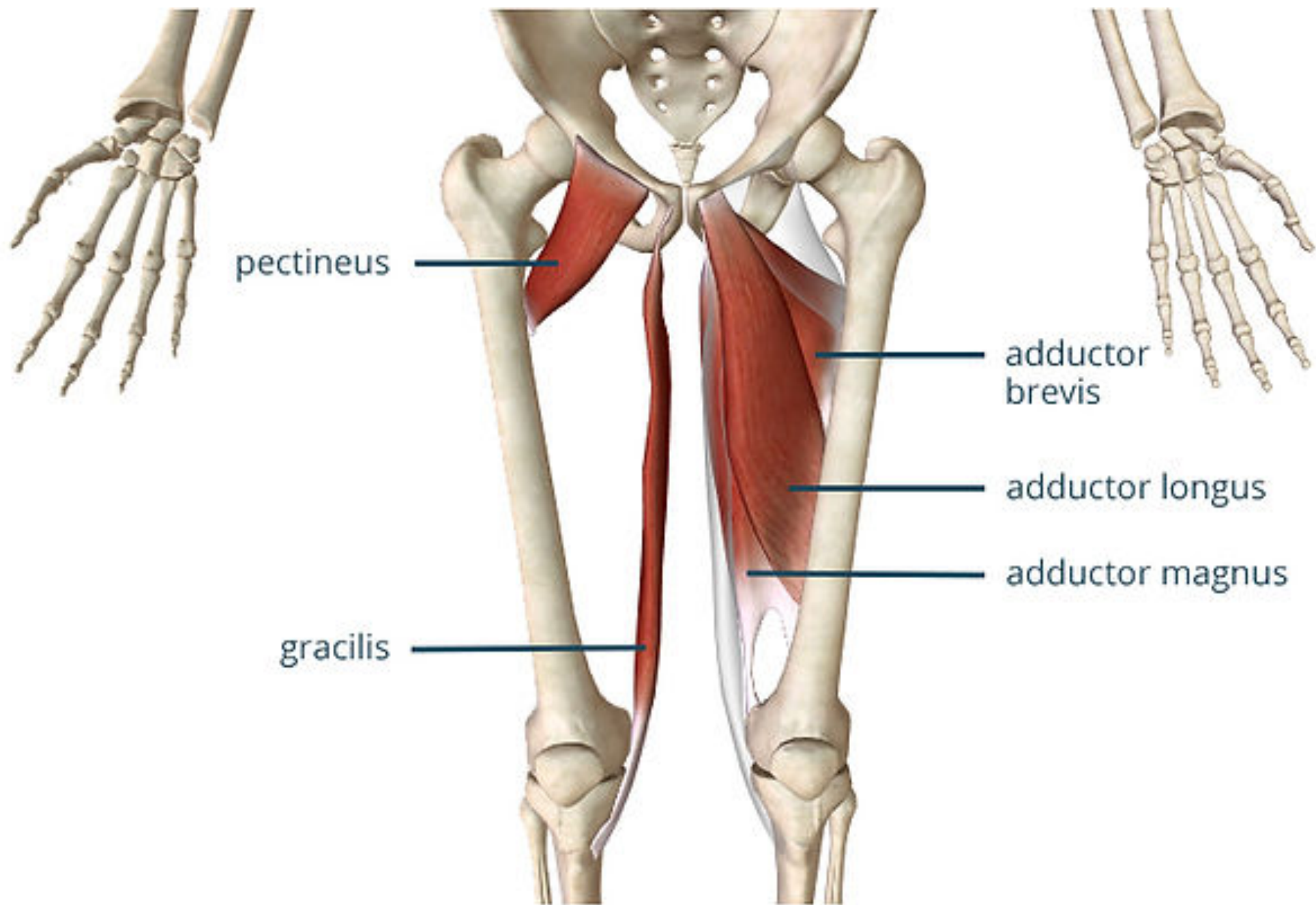
O: from the front of **the body of pubis**, inferior to pubic crest.

I : fibers pass downward & laterally & attached to middle 1/3 of Linea aspera of femur.

N : Obturator nerve

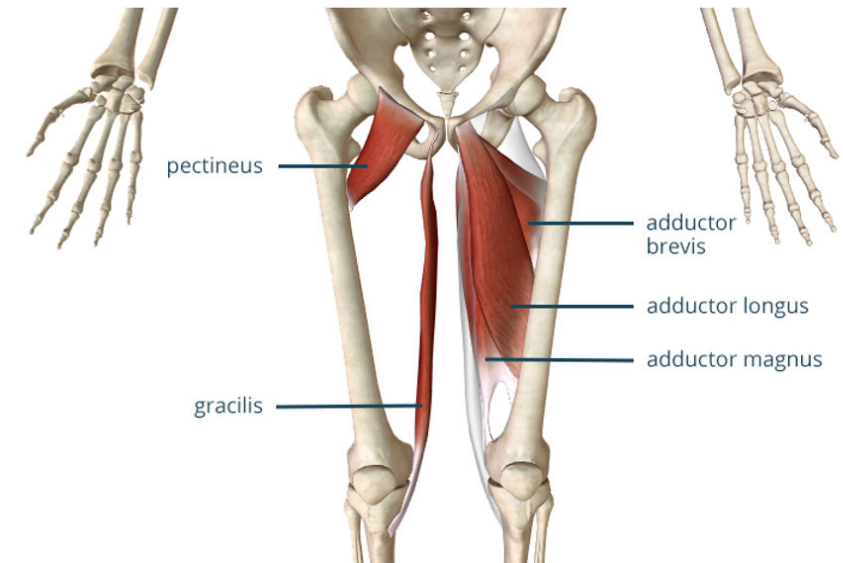
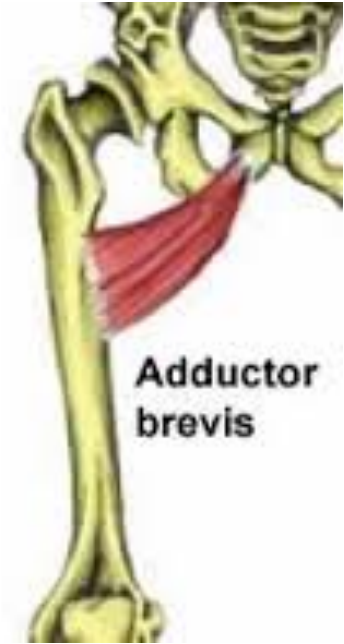
A: **adducts** the thigh at the hip joint & assists in lateral rotation.





3) Adductor brevis

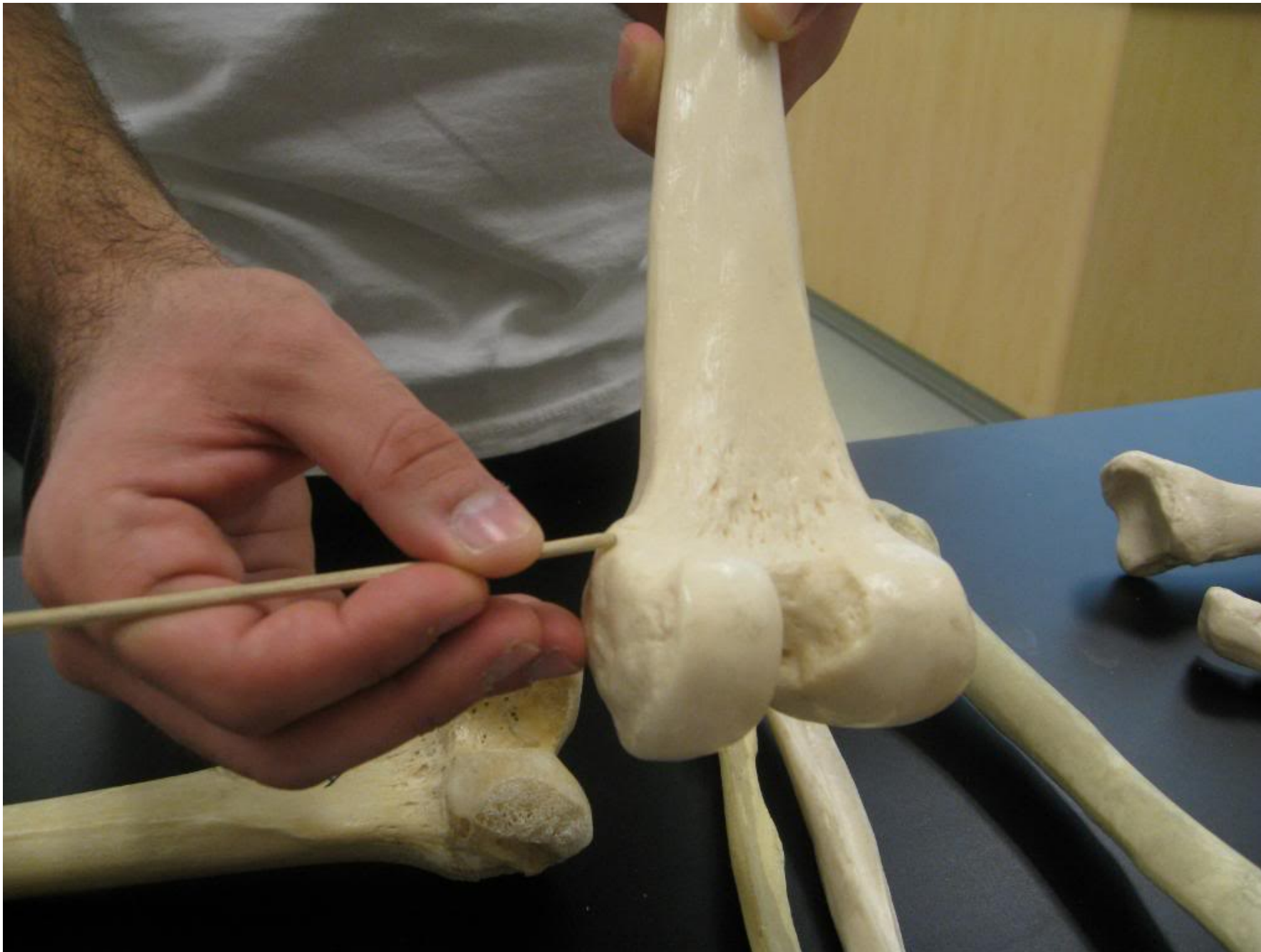
- lies post. to pectineus & adductor longus.
- O:** from outer surface of **inferior ramus of the pubis**.
- I :** fibers pass downward & laterally & attached to **Linea aspera**.
- N :** Obturator nerve
- A:** **Adducts** the thigh at the hip joint & assists in lateral rotation.



4) Adductor magnus

- large, triangular muscle, deepest of 3 adductors.
- O: Adductor portion:** from outer surface of inferior ramus of pubis .
hamstring (ischial) portion: from ramus of ischium & ischial tuberosity .
- I : Adductor portion:** to gluteal tuberosity, linea aspera.
hamstring portion: descends vertically to the adductor tubercle on medial condyle of the femur.
- N: Adductor portion:** by obturator nerve .
Hamstring portion: Tibial part of sciatic nerve
(double(dual) nerve supply)!!
- A: adductor portion:** Adduction & flexion of thigh at hip joint. & assists lateral rotation.
hamstring portion : Extends the thigh at the hip joint.

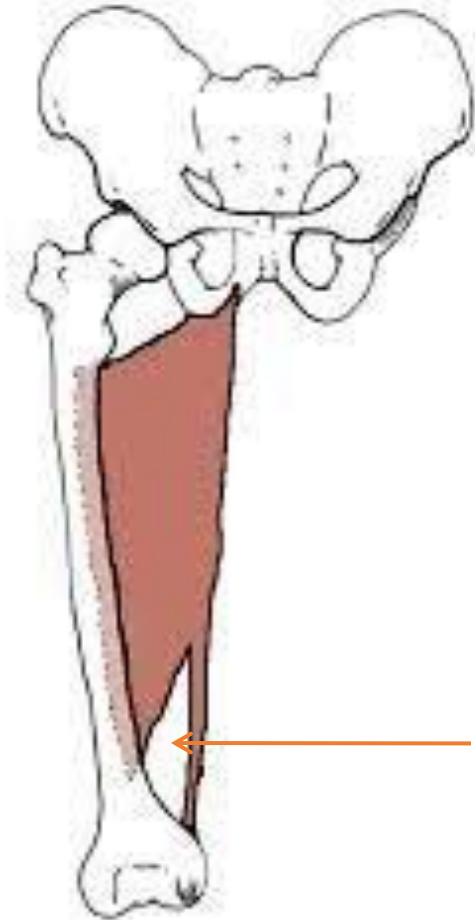
	Adductor portion	hamstring portion
1-Origin	inf.ramus of pubis	ischial ramus & tuberosity
2- insertion	Gluteal tuberosity, linea aspera.	to the adductor tubercle on the medial epicondyle of femur.
3-n.supply	Obturator nerve	Sciatic nerve.
4-Action	Adduction & flexion of thigh at hip joint & assists in lateral rotation.	Extends the thigh at the hip joint.



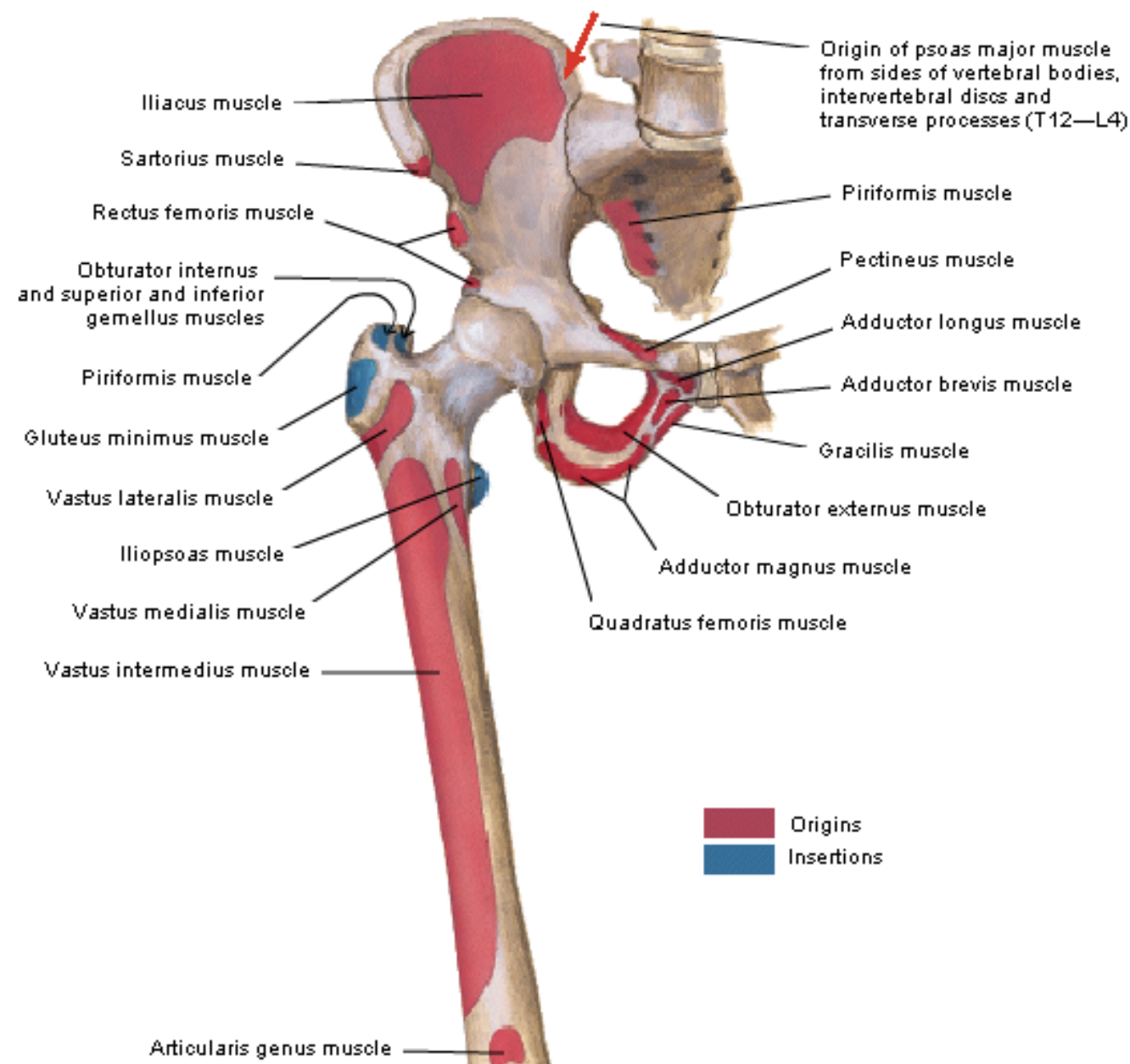
**Adductor
tubercle**

❑ Adductor hiatus

- ❑ A gap in the attachment (or tendon) of two portions of **adductor magnus** to the femur.
- ❑ **Site?** Lat& sup to adductor tubercle.
- ❑ **Importance:** permits femoral **vessels** to pass from the **adductor canal** downward into **popliteal fossa**.

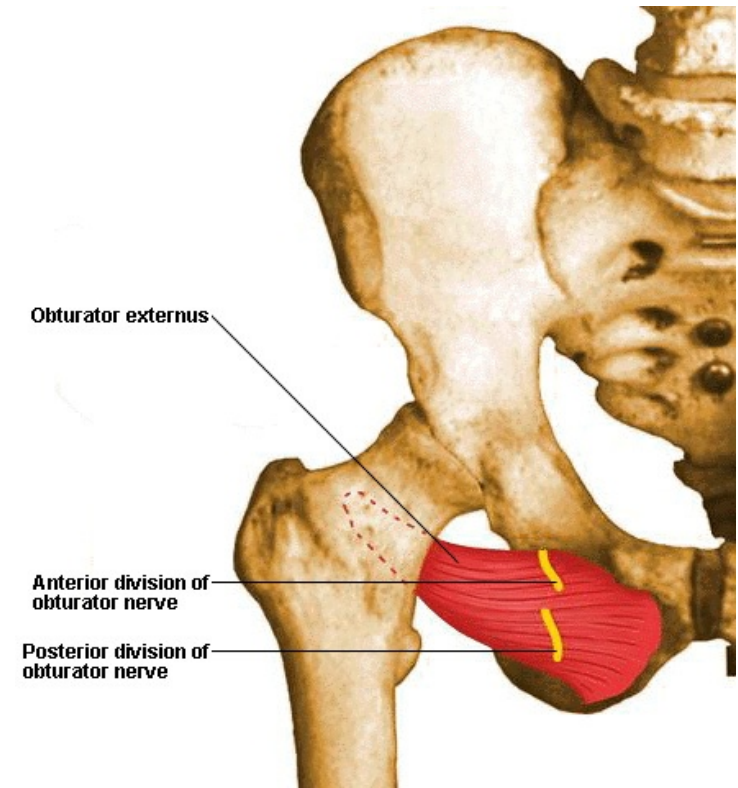


Attachments of Muscles of Hip and Thigh: Anterior View



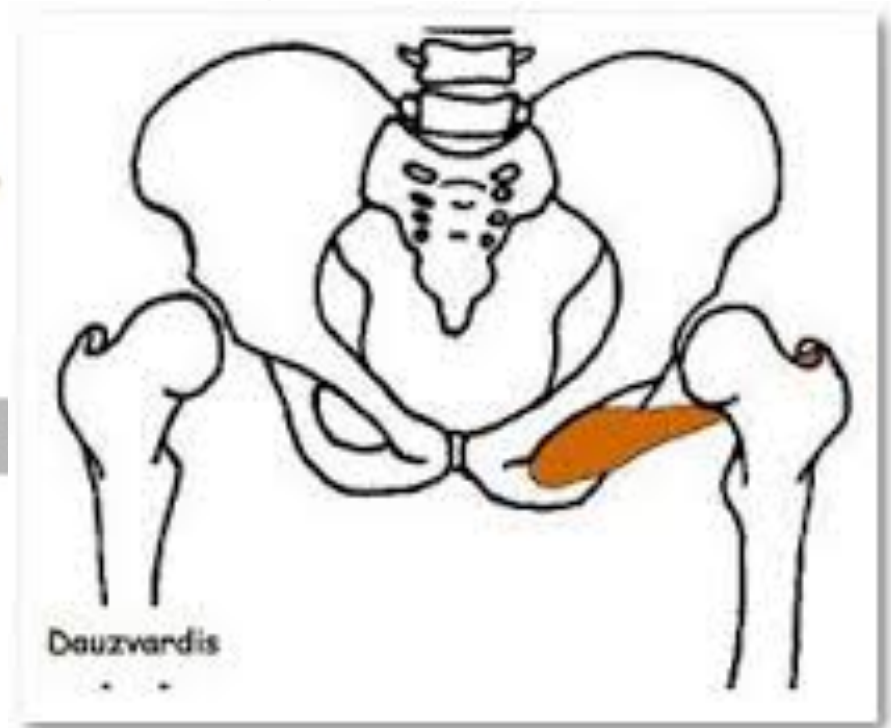
5)Obturator externus

- Small triangular **deeply placed** in superomedial part.
- O** : from outer surface of obturator foramen & obturator membrane.& ischiopubic ramus.
- I** : fibers pass laterally at first
below& then behind the hip joint
to be inserted into **medial side of greater trochanter**.
- N** : Obturator nerve
- A** : **Lateral rotation of thigh** at hip joint.
(Lateral rotator).



Obturator externus

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Anterior Cadaver Thigh Muscles

1. Sartorius
2. Gracilis
3. Adductor magnus
4. Adductor longus



muscle	hip flexion
Iliopsoas	✓
pectineus	✓
Rectus femoris	✓
sartorius	✓
gracilis	✓
Adductor part of adductor magnus	✓



Where is the site of adductor hiatus?

- a. Tendon of adductor brevis
- b. Tendon of adductor longus
- c. Tendon of Adductor magnus
- d. Tendon of sartorius

Which of the following muscles causes flexion of the hip and extension of the knee?

- a. Gracilis
- b. Rectus femoris
- c. Sartorius
- d. Adductor magnus

- What is the nerve supply of gracilis?

- a. Femoral nerve

- b. Obturator nerve

- c. Superior gluteal nerve

- d. Tibial nerve

- Which of the following muscles have double nerve supply?

- a. Pectineus

- b. Adductor magnus

- c. Adductor brevis

- d. a& b



THANK YOU



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